MIL-C-43708B
30 September 1986
SUPERSEDING
MIL-C-43708A
30 December 1981

MILITARY SPECIFICATION

CLOTH, COATED OR CLOTH, LAMINATED, REFLECTIVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers retroreflective materials.

1.2 Classification. The reflective cloth shall be of the following types and classes as specified (see 6.2):

Type I - Cloth, coated, reflectorized coating
Class 1 - Gray color, heavyweight, woven fabric backing
Class 2 - Silver color, lightweight, woven fabric backing
Class 3 - Fluorescent lime-yellow color, knit fabric backing

Type II - Cloth, bonded, reflectorized film
Class 1 - White color
Class 2 - Fluorescent lime-yellow color

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8305

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SPECIFICATIONS

FEDERAL

L-P-375	- Plastic Film, Flexible, Vinyl Chloride
CCC-C-426	- Cloth, Drill, Cotton
PPP-P-1136	- Packaging of Coated (Plastic; Rubber) and Laminated Fabrics

MILITARY

MIL-C-43006	- Cloth and Strip, Laminated or Coated, Vinyl-Nylon or Polyester High Strength, Flexible

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods
FED-STD-601	- Rubber: Sampling and Testing

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1424 - Test Method for Tear Resistance of Woven Fabrics by Falling-Pendulum (Elmendorf) Apparatus

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Samples.

3.1.1 Standard sample. The reflectorized coated or bonded cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.1.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.4).

3.2 Materials. Recycled material may be used (see 6.6).

3.2.1 Type I cloths (base cloths).

3.2.1.1 Class 1. The base cloth shall be cotton drill conforming to type III, class 3 of CCC-C-426 except that colorfastness and preshrinking requirements shall not apply. The color of the base cloth shall be natural or a suitable shade of gray that will produce a reflectorized cloth meeting the color requirements in 3.5.

* 3.2.1.2 Class 2. The base cloth shall conform to the following requirements, when tested as specified in 4.4.1.1.

a. Yarn shall be 65 ± 5 percent polyester and the remainder shall be cotton.
b. Minimum texture shall be 126 yarns per inch for warp and 63 yarns per inch for filling.

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c. Minimum weight shall be 2.8 oz/yd^2.
d. Weave shall be plain (1 x 1).
e. Color shall be natural.

3.2.1.3 Class 3. The base cloth shall be 4.0 oz/yd^2 (minimum), nylon or polyester, tricot knit cloth, natural color. Testing shall be as specified in 4.4.1.1.

3.2.2 Binder. The binder shall be a resin composition that will securely bind the retroreflective glass beads to the base cloth (see 4.4.1.1).

3.2.3 Retroreflective glass beads. The retroreflective glass beads shall be such that the finished coated cloth will meet the applicable reflectivity requirements specified in 3.4.

3.2.4 Coated cloths. The retroreflective glass beads and resin binder shall be applied only to the back side of the base cloth. The finished coated cloth shall conform to the physical requirements specified in table I when tested as specified in 4.4.3, and shall meet the applicable reflectivity requirements specified in 3.4.

3.3 Type II cloths (base cloth and film).

3.3.1 Backing cloth. The backing cloth shall be a vinyl coated or laminated cloth conforming to type II, class 2, form 1 of MIL-C-43006. The color of the coated cloth shall be white.

3.3.2 Reflectorized film. The reflectorized film shall be a vinyl chloride film, of 0.012 inch nominal thickness, conforming to type I, class 1 of L-P-375. The color of the film shall be selected to meet the color requirement specified in 3.5, and the film shall be translucent. The finish on the face side of the film shall be smooth and polished, and the back side shall be embossed or finished such that the finished, bonded cloth will meet the applicable reflectivity requirements specified in 3.4.

* 3.3.3 Bonded cloth. The back side of the reflectorized film shall be bonded to the face side of the backing cloth by means of continuous thermal or dielectric bonds, 1/16 ± 1/32 inch wide, parallel and perpendicular to the warp direction of the cloth. Each edge of the cloth shall be continuously bonded with a bond 3/16 ± 1/16 inch wide. The number and location of the bonds shall be such that the reflectorized film is securely fastened to the backing cloth and that the finished cloth conforms to the applicable reflectivity requirements specified in 3.4. When the cloth is cut into tape or strip form, the bonding shall be such that the edges of the tape shall be continuously bonded with a bond 3/16 ± 1/16 inch, and that the other bonds parallel to the warp direction of the tape shall be located symmetrically

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about the center, warpwise line of the tape. The finished cloth shall conform to the physical requirements specified in table I when tested as specified in 4.4.3.

TABLE I. Physical requirements

Characteristic	Type I						Type II	
	Class 1		Class 2		Class 3		Classes 1 and 2	
	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
Weight, overall (ounces per square yard)	12.5	13.5	8.0	11.0	13.0	15.0	19.0	21.0
Breaking strength (pounds):								
Warp	75	-	50	-	50	-	50	-
Filling	32	-	20	-	40	-	40	-
Tearing strength (grams):								
Warp	-	-	650	-	-	-	-	-
Filling	-	-	350	-	-	-	-	-
Colorfastness to:								
Laundering	1/		-	-	-	-	-	-
Wet dry cleaning	1/		-	-	-	-	-	-
Weather	1/		-	-	-	-	-	-
Light	-		2/		-	-	-	-
Stiffness, machine or warp direction only (centimeters):								
At 70° ±2°F	-	-	-	7.0	-	16.0	-	16.0
At minus 10° ±2°F	-	-	-	12.0	-	-	-	-
Blocking, scale rating	-	-	-	No. 3	-	No. 2	-	No.2
Resistance to cold crack, at minus 20° ±5°F	-	-	-	-	3/		3/	

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TABLE I. Physical requirements - Continued

Characteristic	Type I Class 1 Min.	Type I Class 1 Max.	Type I Class 2 Min.	Type I Class 2 Max.	Type I Class 3 Min.	Type I Class 3 Max.	Type II Classes 1 and 2 Min.	Type II Classes 1 and 2 Max.
Adhesion of coating (pounds per 2-inch width)	-	-	10.0	-	-	-	-	-
Flame resistance (seconds):								
After flame:								
Warp or machine	-	-	-	-	-	2.0	-	2.0
Filling or transverse	-	-	-		-	2.0	-	2.0
After glow:								
Warp or machine	-	-	-	-	-	6.0	-	6.0
Filling or transverse	-	-	-	-	-	6.0	-	6.0
Char length (inches):								
Warp or machine	-	-	-	-	-	4.0	-	4.0
Filling or transverse	-	-	-	-	-	4.0	-	4.0

1/ The coated cloth shall show fastness equal to or better than the standard sample. When no standard sample is available, the cloth shall show "good" fastness.

2/ The coated cloth shall show fastness to light equal to or better than the standard sample after 100 hours exposure when tested as specified in 4.4.3. When no standard sample has been established or designated as applicable to colorfastness, the finished coated cloth shall show "good" fastness to light after 100 hours exposure when tested as specified in 4.4.3.

3/ The coated or bonded cloth shall show no evidence of cracking, flaking, or separation of the coating of the coated cloth or of the components of the bonded cloth.

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3.4 Reflectivity. The reflectivity requirements of the types I and II cloths (see 6.5) shall be as specified in tables II and III when measured at the indicated observation angles and entrance angles (see 4.5.1).

TABLE II. Reflectivity requirements of reflectorized coated cloths

Type	Class	Observation angle (degrees)	Reflectivity (candle power/foot candle/sq ft), minimum at entrance angles (degrees)					
			-4	+10	+20	+30	+40	+50
I	1	0.5	180	140	140	135	120	80
		2.0	21	30	20	20	14	11
I	2	0.2	330	-	290	180	65	-
		0.5	115	-	90	70	20	-
		2.0	4.5	-	4.0	3.5	2.5	-
I	3	0.2	130	-	65	23	6	-
		0.5	20	-	18	15	4.5	-
		2.0	1.5	-	1.2	0.8	0.4	-

TABLE III. Reflectivity requirements of reflectorized bonded film cloths

Type	Class	Observation angle (degrees)	Reflectivity (candle power/foot candle/sq ft), minimum at entrance angles (degrees)		
			-4	+15	+30
II	1	0.2	110	90	40
		0.5	80	75	30
		2.0	6	5	4
II	2	0.2	83	68	30
		0.5	60	56	23
		2.0	4.5	3.8	3.0

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3.5 Color. The color of the coated or bonded cloth shall be as follows and shall match the standard sample (see 6.3):

a. Type I:

Class 1 - Reflective gray 205
Class 2 - Reflective silver
Class 3 - Reflective, fluorescent lime-yellow

b. Type II:

Class 1 - Reflective white
Class 2 - Reflective, fluorescent lime-yellow

* 3.5.1 Matching. The color and shade shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7000 ±500K, with illumination of 100 ± 20 foot candles. It shall be a good match to the standard sample under incandescent lamplight at 2300 ±100K.

3.6 Width. The minimum width shall be specified in the contract (see 6.2). For the coated cloth, the width requirement shall apply to the coated cloth without selvages. Trimming of selvages shall result in straight cut edges, with no rolling or folding occurring. For the bonded cloth, the width shall apply to the bonded cloth or strip with selvages 3/16 ± 1/16 inch wide.

3.7 Length and put-up. The coated or bonded cloth shall be put-up in rolls as specified in 5.1. For the type I cloth, the coated side shall form the inside of the roll. For the type II cloth, the backing side of the cloth shall form the inside of the roll. The rolls shall be not less than 40 yards nor more than 200 yards in length. The maximum number of pieces per roll shall be two, and no single piece shall be less than 6 yards. The ends of the pieces shall be overlapped, not joined by a splice or seam.

3.8 Workmanship. The finished cloth shall be clean, evenly coated or bonded and shall conform to the quality of product established by this document, and the occurrence of defects shall not exceed the acceptable quality levels.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.2.1 and 4.4.2.2 and shall be tested for characteristics specified in 4.4.3. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

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4.4.1.1 Component testing. In addition to the quality assurance provisions of applicable subsidiary documents, testing shall be performed on the base cloths listed in table IV for the characteristics shown. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table IV shall be followed. The lot size shall be expressed in units of yards. The sample unit for the base cloths shall be 1/4 yard full width. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. The sample size shall be as follows:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE IV. Component tests

Component/characteristic	Requirement paragraph	Test method
Base cloth (type I, class 2)		
Polyester identification	3.2.1.2	1/
Cotton identification	3.2.1.2	1200 1/
Fiber content	3.2.1.2	2535 2/
Yarns per inch	3.2.1.2	5050
Weight	3.2.1.2	5041
Weave	3.2.1.2	Visual 3/
Color	3.2.1.2	1/
Base cloth (type I, class 3)		
Fiber identification	3.2.1.3	1/
Weight	3.2.1.3	5041
Type of knit	3.2.1.3	1/
Color	3.2.1.3	1/
Binder	3.2.2	4/

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

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2/ The cotton content shall be calculated as follows:

$$\text{Cotton content in percent} = \left(1 - \frac{R}{S}\right) \times 100,$$

where R = Residue weight (polyester fiber) and
S = Specimen weight.

3/ One determination shall be made per sample unit and the results reported as "pass" or "fail".

4/ A certificate of compliance shall be furnished with each shipment or lot stating that the binder is a resin composition.

4.4.2 End item examination.

4.4.2.1 Yard by yard examination. The required yardage of each roll shall be examined on the reflectorized side only for the defects listed in table V. The defects found shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The lot size shall be expressed in units of yards. The sample unit shall be 1 linear yard. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 4.0 for total defects. The number of rolls from which the sample yardage is to be selected shall be in accordance with table VI. The sample yardage shall be apportioned equally among the selected rolls.

TABLE V. Yard-by-yard examination defects

Defect	Classification	
	Major	Minor
Any cut, hole, tear, scratch, or abrasion mark	X	
Any uncoated area	X	
Any cracked coating	X	
Any heavily coated area		X

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TABLE V. Yard-by-yard examination defects - Continued

Defect	Classification	
	Major	Minor
Any thinly coated area		X
Width less than minimum specified		X
Retroreflective glass beads and binder applied to face side of base cloth (type I only)	X	
Any spot, stain, or streak	X	
Any foreign matter which on removal leaves an uncoated or poorly coated area	X	
Ripples, waviness, or dimensional distortion	X	
Selvages not removed on the type I coated cloth, edges not cut straight, folded or rolled		X
Bonded edges of the type II bonded cloth, strip or tape less than 1/8 inch or more than 1/4 inch in width		X
Color does not match the standard sample	X	
Face side of reflectorized film bonded to face side of backing cloth (type II only)	X	
Back side of reflectorized film bonded to back side of backing cloth (type II only)	X	
Reflectorized film not securely fastened to backing cloth (type II only)	X	
Any roll not having the coated side of the cloth facing the inside of the roll	X	

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TABLE VI. Sample size

Lot size (yards)	Sample size (rolls)	Acceptance number 2/
Up to 1,200 inclusive 1/	3	0
1,201 to 3,200 inclusive	5	0
3,201 to 10,000 inclusive	8	0
10,001 to 35,000 inclusive	13	0
35,001 to 150,000 inclusive	20	1
150,001 and over	32	2

1/ If a lot contains fewer than three rolls, each roll in the lot shall be examined.
2/ Applicable to length examination only (see 4.4.2.2).

* 4.4.2.2 Length examination. Each roll of cloth used in the yard-by-yard examination shall be examined for the defects listed below. If the total number of defects in the sample rolls exceeds the applicable acceptance number specified in table VI or if the total of the actual lengths of the sample rolls is less than the total of the gross lengths marked on the roll tickets, the lot shall be rejected.

Defects:

a. Minor - Any roll gross length less than minimum or more than maximum specified.
b. Major - Any roll gross length more than 2 yards less than gross length marked on ticket.
c. Major - Any splices or seam in any roll.
d. Major - Any roll containing more than two pieces.
e. Major - Any piece in roll less than 6 yards.

* 4.4.3 End item testing. The cloth shall be tested for the characteristics listed in table VII. The methods of testing specified in FED-STD-191, whenever applicable, and as listed in table VII shall be followed. Except for blocking and resistance to cold crack tests, the physical values specified apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. Requirements for blocking and resistance to cold crack shall apply to each determination

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separately. When cloth is being procured as strips, testing shall be performed on the cloth prior to cutting into strip form. The lot size shall be expressed in units of linear yards. The sample unit shall be 3 continuous yards, full width, of the coated or bonded cloth. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. All test reports shall contain the individual values used in expressing the final result. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE VII. End item tests

Characteristic	Requirement paragraph	Test method
Weight (overall)	3.2.4 and 3.3.3	5041
Breaking strength	3.2.4 and 3.3.3	1/
Tearing strength (type I, class 2 only)	3.2.4	ASTM D 1424
Colorfastness to:		
Laundering (type I, class 1 only)	3.2.4	5610
Wet dry cleaning (type I, class 1 only)	3.2.4	5622
Weather (type I, class 1 only)	3.2.4	5671 2/
Light (type I, class 2 only)	3.2.4	5660 2/
Stiffness:		
At 70° ± 2°F	3.2.4 and 3.3.3	5204
At minus 10° ± 2°F (type I, class 2 only)	3.2.4	5204 3/

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TABLE VII. End item tests - Continued

Characteristic	Requirement paragraph	Test method
Blocking	3.2.4 and 3.3.3	5872
Resistance to cold crack at minus 20° ± 5°F	3.2.4 and 3.3.3	5874 4/
Adhesion of coating (type I, class 2 only)	3.2.4	5970
Flame resistance (type I, class 3 and type II, classes 1 and 2 only)	3.2.4 and 3.3.3	5903
Reflectivity	3.4	4.5.1

1/ For type I, class 1 coated cloth, Method 5100 shall be used; for all other types and classes, Method 5102 shall be used.

2/ The time of exposure shall be 100 hours.

3/ The test specimen shall be subjected to a temperature of minus 10° ± 2°F for a minimum of 4 hours, and the test shall be performed in a still atmosphere at that temperature.

4/ The test specimens shall be subjected to a temperature of minus 20° ± 5°F for not less than 4 hours before testing. The test for hydrostatic resistance shall not be performed.

* 4.4.3.1 Standard test conditions. Results of physical tests obtained under testing conditions defined in FED-STD-191 or FED-STD-601 will be acceptable except in case of dispute. In disputed cases, tests shall be conducted with both the specimen and test apparatus under standard conditions as defined in FED-STD-191.

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* 4.4.4 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of PPP-P-1136.

4.5 Method of inspection.

4.5.1 Reflective intensity.

4.5.1.1 Apparatus. Arrangement of the apparatus for the retroreflectivity test shall be as shown in figure 1. A light projector having a lens diameter of 1 to 3 inches and capable of projecting a uniform field of light shall be used to illuminate the specimens. The source of illumination shall be a 750 watt projection lamp having a color temperature in the range of 2850 to 3250K. The level of illumination at the specimen surface as measured with a light meter shall be 4 to 6 foot candles (3.75 to 5 foot lamberts). The light reflected from the test specimen surface (specimen brightness) shall be measured using a photoelectric receiver whose response can be read in foot lamberts either directly or from a calibration chart. The photoelectric receiver shall be a cadmium sulphide cell or one having reponse characteristics comparable to that of cadmium sulphide. The photoelectric receiver shall be capable of reading accurately to 0.01 foot lamberts at the smallest incidence angles (i.e., at the lowest brightness levels). The aperture of the receiver shall be such as to provide an acceptance angle of 1 degree and an overall viewing angle of 21 degrees. The active area of the receiver shall be mounted normal to the beam of retroreflected light. The surface for mounting the test specimen shall be a black nonreflecting surface larger than the specimen to be tested.

4.5.1.2 Specimen. The specimen shall be not less than 1 foot square and shall be mounted in an 18-inch by 18-inch holder in nonreflective black color. The holder shall be mounted on a turntable or similar device to provide the required angles of incidence as indicated in 3.4.

4.5.1.3 Procedure. With the test apparatus set in a darkened room, mount the specimen on the black mounting surface so that the surface of the specimen being tested shall be 50 feet ± 2 inches from the projector lens and the receiver. Turn on the projector and adjust it so that the incident light falls entirely on the specimen surface. Measure the illumination incident on the receiver due to reflection from the test surface at six angles of incidence for each of the two angles of divergence as specified in 3.4. Read the reflected brightness expressed in light level values directly from the light meter at each angle of incidence for each angle of divergence. Take the average of three determinations, and convert this average value to reflective brightness in foot lamberts by the use of tables or charts supplied with the reflective brightness meter.

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5. PACKAGING

5.1 Put-up and preservation. Put-up and preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Levels A and Commercial. The coated cloth shall be put-up and preserved in accordance with the applicable requirements of PPP-P-1136.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Levels A, B, and Commercial. The coated cloth shall be packed in accordance with the applicable requirements of PPP-P-1136.

5.3 Marking. In addition to any special marking required by the contract or purchase order, unit packs and shipping containers shall be marked in accordance with PPP-P-1136.

6. NOTES

6.1 Intended use. The reflectorized coated and bonded cloths are intended for use in the manufacture of Safety Clothing, High Visibility (MIL-S-43753); Coat, Fireman's (MIL-C-10750); and Trousers, Fireman's (MIL-T-3901).

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type and class required (see 1.2).
c. When a first article is required (see 3.1.2, 4.3, and 6.4).
d. Width required (see 3.6).
e. Selection of applicable levels of put-up, preservation, and packing (see 5.1 and 5.2).

6.3 Standard sample. Standard color samples of the reflectorized cloth may be obtained from the contracting activity (see 3.1.1 and 3.4). For coated cloth, roll number 1439 may be used as a reference standard for calibration. Roll number 1439 has the following readings when measured with the equipment referenced in 6.5.

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Entrance angle degree	0.5°	2.0°
-4	225	21
+10	182	30
+20	182	21
+30	175	21
+40	160	14
+50	130	11

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample consisting of 2 yards of the finished cloth. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.5 Reflectivity values. The values contained in 3.4 were obtained by measurement of retroreflective materials conforming to this document using the following equipment:

a. Illumination meter - Huygen Corp., Model 615

b. Sample holder - 18-inch by 18-inch frame on turntable calibrated in angles of incidence in accordance with 3.4; receiver positions controlled by extended arm graduated in angles of divergence as given in 3.4.

6.5.1 Conversion from light level readings to foot lamberts. The conversion from light readings on the Pentax Meter to foot lamberts was in accordance with the following table:

Light level	Foot lamberts
3	0.25
4	0.50
5	0.97
6	1.9
7	4.0
8	7.8
9	15.7
10	31.4
11	62.8
12	125
13	251
14	502
15	1004

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Light level	Foot lamberts
16	2009
17	4019
18	8038

6.6 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.2).

* 6.7 Subject term (by key word) listing.

Cloth
Coated cloth
Laminated cloth
Reflective cloth
Safety clothing

6.8 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Air Force - 11

Review activities:

Army - MD
Air Force - 82, 99
DLA - CT

Preparing activity:

Army - GL

Project No. 8305-0080

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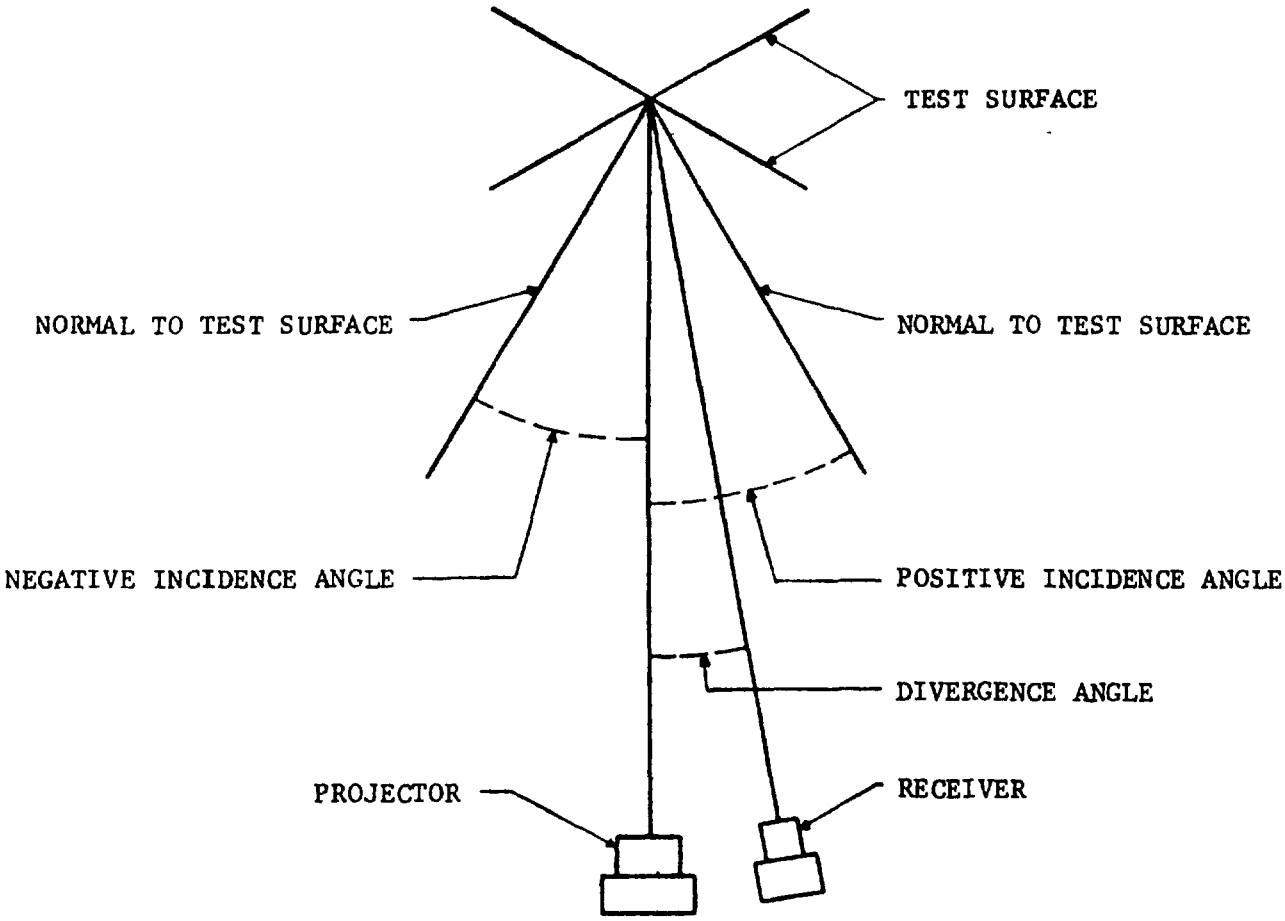


Figure 1. Incidence and Divergence Angles

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-43708 B	CLOTH, COATED GLCLOTH, LAMINATED REFLECTIVE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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